

The W4 method: A new multi-dimensional root-finding scheme for nonlinear systems of equations

Hirotsada Okawa^{a,*}, Kotaro Fujisawa^b, Yu Yamamoto^c, Ryosuke Hirai^{d,e},
Nobutoshi Yasutake^{f,g}, Hiroki Nagakura^h, Shoichi Yamada^c

^a Waseda Institute for Advanced Study, Waseda University, Tokyo 169-0051, Japan

^b Department of Physics, Graduate School of Science, University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan

^c Research Institute for Science and Engineering, Waseda University, Tokyo 169-8555, Japan

^d OzGrav: Australian Research Council Centre of Excellence for Gravitational Wave Discovery, Clayton, VIC 3800, Australia

^e School of Physics and Astronomy, Monash University, Clayton, Victoria 3800, Australia

^f Physics Department, Chiba Institute of Technology, Chiba 275-0023, Japan

^g Advanced Science Research Center, Japan Atomic Energy Agency, Tokai, Ibaraki 319-1195, Japan

^h Division of Science, National Astronomical Observatory of Japan, 2-21-1 Osawa, Mitaka, Tokyo 181-8588, Japan

ARTICLE INFO

Article history:

Received 28 April 2022

Received in revised form 22 August 2022

Accepted 27 August 2022

Available online 5 September 2022

Keywords:

System of nonlinear equations

Iterative methods

Partial differential equations

ABSTRACT

We propose a new class of method for solving nonlinear systems of equations, which, among other things, has four nice features: (i) it is inspired by the mathematical property of damped oscillators, (ii) it can be regarded as a simple extension to the Newton-Raphson (NR) method, (iii) it has the same local convergence as the NR method does, (iv) it has a significantly wider convergence region or the global convergence than that of the NR method. In this article, we present the evidence of these properties, applying our new method to some examples and comparing it with the NR method.

© 2022 The Authors. Published by Elsevier B.V. on behalf of IMACS. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

One of the most important problems in computational science and engineering is the root-finding of functions. It is defined as a problem to solve the following equations numerically:

$$\mathbf{F}(\mathbf{x}) = 0, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^N$, $N \in \mathbb{Z}$ and $\mathbf{F} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a generic system of nonlinear equations in $\mathbf{x}$.

Numerical root-finding methods are essentially categorized into two types. One is deterministic methods such as the Newton-Raphson (NR) method [9], which always give the same answer to the same initial condition. The other is stochastic methods such as the Monte-Carlo method [8], in which the answer may change in each trial even for the identical initial conditions. The former method normally requires a larger computational cost but needs a very small number (mostly one) of trials, whereas the computational cost per trial is small but many trials are needed in the stochastic method.

Although the single-variable problem is rather simple, it becomes highly nontrivial when the dimension of variable space is higher than one. One may choose the NR method if the dimension is not prohibitively large in terms of numerical cost.

* Corresponding author.

E-mail address: h.okawa@aoni.waseda.jp (H. Okawa).

The NR method can be regarded as an iterative solver based on the fixed point theorem and it is guaranteed to give a solution as long as the initial guess is sufficiently close to the solution [13]. In practice, there are two major drawbacks in the NR method: (i) very heavy computational cost in the inversion of a large Jacobian matrix (see Eq. (6)) and (ii) the strong dependence of convergence on the initial guess. The former is evaluated as $\mathcal{O}(N^3)$ for the direct inversion of an $N \times N$ Jacobian matrix in the system of N -dimensions. So far, much attention has been paid to issue (i) and many efforts have been made to reduce the cost: for example, a number of quasi-Newton methods have been proposed that successfully reduce the computational costs to $\mathcal{O}(N^2)$ [1,9,18,3]. As for issue (ii), many studies have attempted to accelerate the normally quadratic convergence of the NR method. In fact, Halley's and Householder's methods proposed for single-variable problems are headed in this direction [6,2,11]. Also for multi-variable problems, a new approach emerged recently [16,15]. See, e.g., [17] for these and other recent progresses with some applications. One of the important targets we have in mind for the application of such root-finding methods is astrophysics problems. For example, the structure of a massive compact star is described by the Einstein equation and Euler equation. Due to their highly nonlinear natures, numerical solutions of the Einstein equation for gravitational fields and the Euler equation for gas are complicated and require a reliable root finder frequently. It is well known, however, that the NR method is not perfect [14].

It is the sensitivity of convergence to the initial guess that we focus most in this paper. It is also well known that the NR method fails to find solutions when the initial guess is far from the solution, in particular, in multi-dimensional problems. A serious consideration needs to be given to the initial guess but there is no guarantee that it is always successful unless the solution is known. When the NR method fails to find solutions, there is either a divergence or a permanent oscillation in the successive solutions of the linearized equations as explicitly shown in Sec. 2. This is sometimes related to the existence of singularities in the Jacobian matrix and may be solved by modifying source terms in the linearized equations [10,23,7]. It should also be emphasized that the inversion of ill-conditioned matrices usually need high-precision calculations regardless of the iterative methods employed and a careless handling could easily produce an incorrect inverse matrix whose component is wrong from the leading figure. In such cases, numerical errors accumulated during iterations should be inspected in detail. In this article, we focus on problems, for which the NR method fails to find solutions in spite of the fact that its Jacobian matrix is well-conditioned initially at least. The initial guess is likely to be a culprit then. We demonstrate that our new method can find a solution even when an initial guess is far from the solution.

This paper is organized as follows. Firstly, in section 2, we raise the issues of our concern in this paper, giving some concrete but simple examples, and make clear the motivation of this work. Then, we present our new formulation to find roots of a given system of nonlinear equations in Sec. 3, starting with the descriptions of other conventional methods in the literature on the same basis. In Sec. 4, we demonstrate the capabilities of our method for several sample problems including those given in section 2, comparing them with those of other methods. Section 5 is dedicated to the application of our new method to a problem with larger dimensions, which may serve as a surrogate to astrophysical problems we have in mind for future applications. Finally, a summary and comments on the future prospects are given briefly in Sec. 6.

2. Issues

As mentioned in Introduction, our goal is to numerically solve generic systems of nonlinear equations, Eq. (1). In order to make clear the issues of our concern in this article, some specific examples are picked up to show the problems with the conventional methods. Although our main interest is in multi-variable equations, the discussion starts with a single-variable problem for clarity.

2.1. A single-variable problem

The root-finding of the following single-variable equation is considered first:

$$f(x) = \arctan(x) + \sin(x) - 1 = 0, \quad (2)$$

where $x \in \mathbb{R}$. The left panel of Fig. 1 displays the function, which has oscillatory behavior and has multiple solutions only on the positive x -axis ($x > 0$). In the NR method, one employs the following iteration map from the n -th step to the $(n+1)$ -th step,

$$x_{n+1} = x_n - \left\{ \frac{df}{dx}(x_n) \right\}^{-1} f(x_n). \quad (3)$$

As demonstrated in Table 1, when the NR method obtains a solution, it requires only a small number of iteration steps to get to the solution within a prescribed accuracy, i.e., $|f(x)| < 10^{-6}$ in this example, thanks to its well known quadratic-convergence nature. The table also demonstrates that the NR method fails to find a solution if the initial guess is smaller than $x_i = -2.0$. The right panel of Fig. 1 shows typical trajectories of the NR map, Eq. (3), both for the successful and failed attempts. The iteration rapidly converges to one of the solutions ($x = 1$) when the initial guess is close to the solution whereas it shows an oscillation between two values and never reaches a solution if the initial guess is $x_i = -2.0$. Since the latter oscillation is a consequence of the oscillatory nature of the function $f(x)$, it is a natural tactic to reduce the slope somewhat by hand in the evaluation of x in the next step as

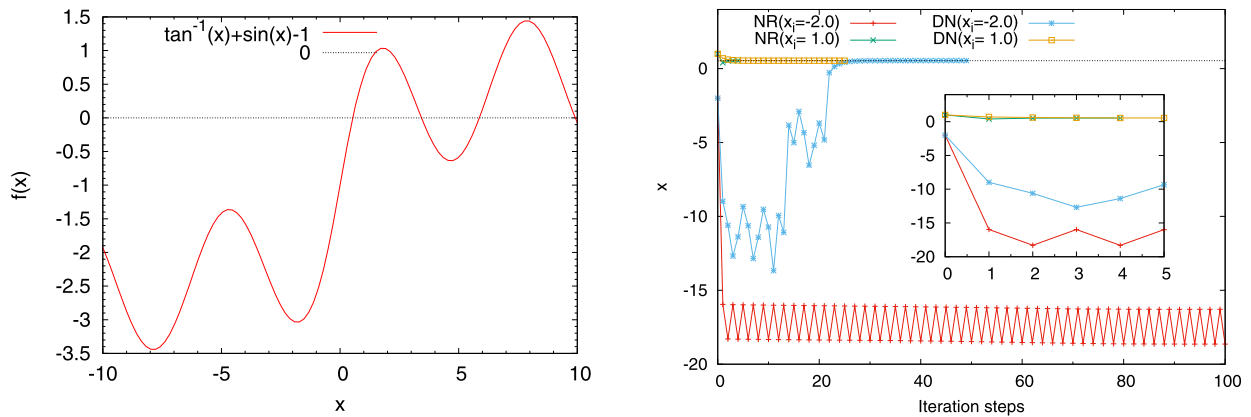


Fig. 1. (Left) The function given in Eq. (2). (Right) Some representative sequences produced by the iteration maps for the NR (blue and purple) and DN (orange and green) methods. The initial conditions are set either to $x_i = 1.0$ or to $x_i = -2.0$. The horizontal dotted line indicates the solution. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 1

The numbers of iteration steps it takes the NR and DN methods to get convergence for different initial guesses x_i . The damping parameter for the DN method is set to $\Delta\tau = 0.5$. The stopping criterion in both methods is $|f(x)| < 10^{-6}$. We put ∞ when no solution is obtained by 10^4 iteration steps.

Method \ x_i	-3.0	-2.5	-2.0	-1.5	-1.0	-0.5	0	0.5	1.0	1.5	2.0	2.5	3.0
NR	∞	∞	∞	4	5	4	3	2	4	8	4	4	3
DN	25	∞	41	20	19	20	19	15	18	19	17	19	18

$$x_{n+1} = x_n - \Delta\tau \left\{ \frac{df}{dx}(x_n) \right\}^{-1} f(x_n), \quad (4)$$

where a damping parameter $\Delta\tau$ is introduced, which is a positive real number less than unity. The results for this so-called Damped Newton (DN) method are presented in Table 1 for the comparison with the original NR method. Although the number of iteration steps required to get a solution within the prescribed accuracy increases substantially in the DN method, the region of the initial guess, where a solution is obtained, becomes wider. This implies that the NR method, if appropriately modified, may find a solution even for initial guesses that are not very close to the solution at least for single-variable problems in principle.

2.2. A multi-variable problem

The above situation is changed drastically when the dimension of variable space becomes higher than one. To observe what happens, the following concrete problem is considered:

$$F_1(x, y) := x^2 + y^2 - 4 = 0, \quad (5a)$$

$$F_2(x, y) := x^2 y - 1 = 0. \quad (5b)$$

The functions F_1 and F_2 represent a circle and a curve, respectively, and the roots of this system correspond to the intersections of these two curves in the $x - y$ plane. In the NR or DN method for multi-variable problems [13,9], one can proceed just as in the single-variable case and employs the following iteration map from the n -th step to the $(n + 1)$ -th step,

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \Delta\tau J^{-1} \mathbf{F}(\mathbf{x}_n), \quad (6)$$

where $\mathbf{F} := (F_1 \ F_2)^T$ and J^{-1} is the inverse of the Jacobian matrix associated with Eqs. (5), which is given as

$$J = \begin{bmatrix} 2x & 2y \\ 2xy & x^2 \end{bmatrix}. \quad (7)$$

Then, the iteration maps for $\mathbf{x} = (x \ y)^T$ are expressed explicitly as

$$x_{n+1} = x_n - \Delta\tau \left[\frac{x_n}{2(x_n^2 - 2y_n^2)} F_1 - \frac{y_n}{x_n(x_n^2 - 2y_n^2)} F_2 \right], \quad (8a)$$

$$y_{n+1} = y_n - \Delta\tau \left[-\frac{y_n}{x_n^2 - 2y_n^2} F_1 + \frac{1}{x_n^2 - 2y_n^2} F_2 \right]. \quad (8b)$$

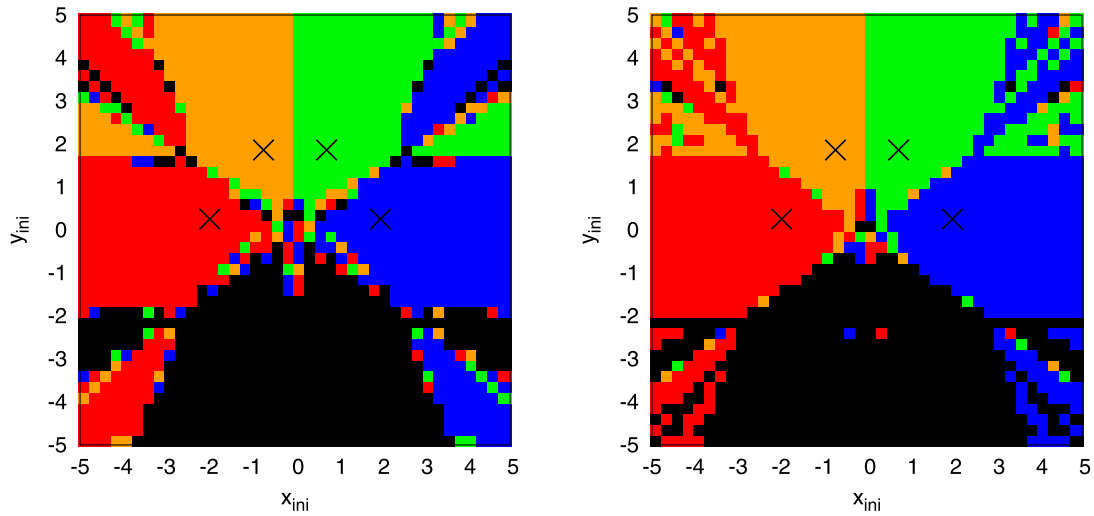


Fig. 2. The initial-guess dependence for the NR method (Left) and the DN method (Right) with $\Delta\tau = 0.5$. The color (red, orange, green or blue) at each point shows one of the four solutions of Eqs. (5) marked with crosses, whereas black means that no solution is reached from that point.

Fig. 2 displays the so-called Newton basin for the NR and DN maps in the left and right panels, respectively, as an indicator of the initial-guess dependence of these methods. Different colors (red, orange, green and blue) correspond to one of the four solutions,

$$(\mathbf{x}^*, y^*) \approx (\pm 1.9837924, 0.25410169), (\pm 0.73307679, 1.8608059), \quad (9)$$

to which each point converges. The black color implies that none of the solutions has been reached within 1000 iteration steps. For both methods, the local convergence is observed more or less as expected. It is evident, however, that the damping factor introduced in the DN method does not help as much in finding solutions in this two dimensional problem as in the one-dimensional counterpart considered in the previous section. Note in particular that the black region, from which no solution can be reached, is not narrowed essentially. This is what we want to change with our new method.

3. The new method

3.1. The NR method as a relaxation

The iteration map given in Eq. (6) can also be regarded as a discretized form of the following continuous time-evolution equation:

$$\frac{d\mathbf{x}}{d\tau} = -J^{-1}\mathbf{F}(\mathbf{x}). \quad (10)$$

A slightly more general time-evolution equation with a preconditioning matrix M may be given as

$$\frac{d\mathbf{x}}{d\tau} = -M\mathbf{F}(\mathbf{x}). \quad (11)$$

The corresponding discretized form is obtained by the Euler forward discretization in time with the time interval $\Delta\tau$ as

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \Delta\tau M\mathbf{F}(\mathbf{x}_n), \quad (12)$$

where $\mathbf{x}_n$ is the discretized solution at the n -th iteration step. If one employs the inverse of the Jacobian matrix as the preconditioner $M = J^{-1}$, then Eq. (6) is recovered.

Theorem 1. Suppose the solution of Eq. (1) exists and that $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is Lipschitz continuous near the solution $\mathbf{x}^*$, then the iteration map (12) for $M = J^{-1}$ with $0 < \Delta\tau < 1$ converges linearly to the solution $\mathbf{x}^*$ starting from an arbitrary initial guess sufficiently close to the solution.

Proof. The Taylor expansion of the source function $\mathbf{F}(\mathbf{x})$ around the solution $\mathbf{x}^*$ up to the first order in the error defined as $\mathbf{e}_n := \mathbf{x}^* - \mathbf{x}_n$ gives

$$\mathbf{F}(\mathbf{x}^*) = \mathbf{F}(\mathbf{x}_n) + J\mathbf{e}_n + \mathcal{O}(\|\mathbf{e}\|^2). \quad (13)$$

Since $\mathbf{x}^*$ is the solution and $\mathbf{F}(\mathbf{x}^*) = 0$, one can express $\mathbf{F}(\mathbf{x}_n)$ in Eq. (12) in terms of the Jacobian matrix and the error to yield the error propagation equation as

$$\mathbf{e}_{n+1} = (I - \Delta\tau MJ)\mathbf{e}_n, \quad (14)$$

where I is the $N \times N$ identity matrix. Under the present condition $M = J^{-1}$ and $0 < \Delta\tau < 1$ the error converges to zero in the $n \rightarrow \infty$ limit,¹

$$\lim_{n \rightarrow \infty} \frac{\|\mathbf{e}_{n+1}\|}{\|\mathbf{e}_0\|} = \lim_{n \rightarrow \infty} (1 - \Delta\tau)^{n+1} = 0. \quad (15)$$

As a result, the iteration map (12) leads to the solution $\mathbf{x}^*$. $\square$

It is again emphasized that this theorem guarantees only the local convergence, i.e. a solution is obtained only when the initial guess is close enough to the solution.

3.2. A new approach for the single-variable problems

Now we proceed to the new approach we propose in this article. We begin with the single-variable problem again. For a nonlinear equation with a single variable

$$F(x) = 0, \quad (16)$$

the following time-evolution equation with the second derivative in time τ is considered:

$$\frac{d^2x}{d\tau^2} + c\frac{dx}{d\tau} + \left\{\frac{dF}{dx}\right\}^{-1} F(x) = 0, \quad (17)$$

which is equivalently expressed with two first-order equations:

$$\frac{dx}{d\tau} := \dot{x} = p, \quad (18a)$$

$$\frac{dp}{d\tau} := \dot{p} = -cp - \left\{\frac{dF}{dx}\right\}^{-1} F(x), \quad (18b)$$

where the dot notation is introduced for the derivative with respect to τ for later convenience and $c \in \mathbb{R}$ is a constant parameter. These are nothing but damped oscillator's equations of motion if $F(x) = x$ and $c > 0$.

Lemma 1. Suppose there exists a solution of the nonlinear equation (16) denoted by x^* . If the parameter is set to $c=2$, then a sufficiently small deviation from x^* always decays during its time evolution (17).

Proof. The Taylor expansion around the solution yields

$$F(x^* + \delta x) = F(x^*) + \frac{dF}{dx}\delta x + \mathcal{O}(\delta x^2). \quad (19)$$

One can then obtain the equation for the deviation δx valid up to the first order by substituting this relation and $x = x^* + \delta x$ into Eq. (17) as

$$\delta\ddot{x} + c\delta\dot{x} + \delta x = 0. \quad (20)$$

There exists a characteristic mode for equation (20), which is explicitly given by assuming $\delta x = e^{i\omega\tau}$ as

$$\omega = \frac{ic \pm \sqrt{-c^2 + 4}}{2}. \quad (21)$$

The deviation from the solution x^* thus decays exponentially as $\delta x = e^{-\tau}$ during the time evolution. $\square$

Theorem 2. (Bendixson's criterion) If the parameter c is nonzero in a simply connected region $\mathcal{D} \subset \mathbb{R}^2$ in the phase space (x, p) , then there are no periodic orbits in $\mathcal{D}$.

¹ In general, the necessary and sufficient condition for the map (12) to converge to a solution is that the absolute values of all the eigenvalues for the matrix $E - \Delta\tau MJ$ should be less than unity.

Proof. Any orbit of system (18) in the two-dimensional phase space (x, p) is described by $dp/dx = \dot{p}/\dot{x}$. If a closed orbit Γ were existent in $\mathcal{D}$, it should satisfy

$$\oint_{\Gamma} (\dot{x} dp - \dot{p} dx) = 0. \quad (22)$$

However, Green's theorem would give

$$\oint_{\Gamma} (\dot{x} dp - \dot{p} dx) = \iint_S \left(\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{p}}{\partial p} \right) dx dp = -c|S| = 0, \quad (23)$$

where S is the region enclosed by Γ and $|S|$ is its area. Consequently, no periodic orbit exists inside $\mathcal{D}$ in the phase space with a nonzero parameter c . $\square$

Lemma 1 ensures the local convergence of Eq. (17) just as in Eq. (11) whereas Theorem 2 may underlie a better global convergence of Eq. (17) that we observe later.²

The explicit Euler finite-differencing in time with the time interval of $\Delta\tau$ leads to the discretized version of equations (18) as

$$x_{n+1} = x_n + \Delta\tau p_n, \quad (24a)$$

$$p_{n+1} = (1 - c\Delta\tau)p_n - \Delta\tau \left\{ \frac{dF}{dx} \right\}^{-1} F(x_n). \quad (24b)$$

The iteration also converges to the same solution under a certain condition.

Proposition 1. Suppose there exists a solution of Eq. (16) and that $F : \mathbb{R} \rightarrow \mathbb{R}$ is Lipschitz continuous near the solution x^* , then the iteration map (24) with $c = 2$ and $0 < \Delta\tau < 1$ yields a series of x_n that converges to the solution x^* starting from $p_0 = 0$ and an arbitrary initial guess x_0 that is sufficiently close to the solution $x = x^*$.

Proof. Just as in the proof of Theorem 1, the error propagation for the map (24) is evaluated up to the first order in $e_n^{(x)} := x^* - x_n$ and $e_n^{(p)} := 0 - p_n$, by using $F(x_n) = -\frac{dF}{dx}e_n^{(x)}$, as follows:

$$e_{n+1}^{(x)} = e_n^{(x)} + \Delta\tau e_n^{(p)}, \quad (25a)$$

$$e_{n+1}^{(p)} = (1 - 2\Delta\tau)e_n^{(p)} - \Delta\tau e_n^{(x)}. \quad (25b)$$

Then the following solution to these equations is obtained

$$e_n^{(x)} = e_0^{(x)} (1 - \Delta\tau)^{n-1} [1 + (n-1)\Delta\tau], \quad (26a)$$

$$e_n^{(p)} = -ne_0^{(x)} \Delta\tau (1 - \Delta\tau)^{n-1}, \quad (26b)$$

which can be easily confirmed by substitution. The convergence is now evident for $0 < \Delta\tau < 1$,

$$\lim_{n \rightarrow \infty} e_n^{(x)} = \lim_{n \rightarrow \infty} e_n^{(p)} = 0. \quad \square \quad (27)$$

3.3. The new formulation for the multi-variable problem

Now we move on to the multi-variable case, the main part of this paper. Unlike the NR method, Eq. (11), discussed in Sec. 3.1, the new method is based on the following second-order time-evolution equations instead of the first-order ones:

$$\frac{d^2 \mathbf{x}}{d\tau^2} + M_1 \frac{d\mathbf{x}}{d\tau} + M_2 \mathbf{F}(\mathbf{x}) = 0, \quad (28)$$

where two preconditioning matrices are introduced as $M_1, M_2 : \mathbb{R}^N \rightarrow \mathbb{R}^N$. These equations are then divided into two sets of first-order differential equations as follows:

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{X}\mathbf{p}, \quad (29a)$$

$$\frac{d\mathbf{p}}{d\tau} = -2\mathbf{p} - \mathbf{Y}\mathbf{F}, \quad (29b)$$

² This analysis will not hold for higher dimensional problems, since chaos may kick in and trajectories will not be closed.

which $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{p} \in \mathbb{R}^N$ are generalized coordinates and associated momenta, respectively; another pair of preconditioners X and Y are introduced, which will be determined later. For the multi-variable case, the constant parameter is always set to $c = 2$. The explicit Euler discretization in time with the time interval $\Delta\tau$ then yields the following iteration map:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta\tau X \mathbf{p}_n, \quad (30a)$$

$$\mathbf{p}_{n+1} = (1 - 2\Delta\tau) \mathbf{p}_n - \Delta\tau Y \mathbf{F}(\mathbf{x}_n). \quad (30b)$$

Again linearizing the above nonlinear map around the solution ($\mathbf{x} = \mathbf{x}^*$ and $\mathbf{p} = \mathbf{0}$), we obtain the following error propagation equations:

$$\mathbf{e}_{n+1}^{(x)} = \mathbf{e}_n^{(x)} + X \Delta\tau \mathbf{e}_n^{(p)}, \quad (31a)$$

$$\mathbf{e}_{n+1}^{(p)} = (1 - 2\Delta\tau) \mathbf{e}_n^{(p)} - Y J \Delta\tau \mathbf{e}_n^{(x)}, \quad (31b)$$

which can be cast into a more compact form as

$$\mathbf{e}_{n+1}^{(z)} = W \mathbf{e}_n^{(z)}, \quad \mathbf{e}_n^{(z)} := \begin{pmatrix} \mathbf{e}_n^{(x)} \\ \mathbf{e}_n^{(p)} \end{pmatrix}, \quad W := \begin{bmatrix} I & \Delta\tau X \\ -\Delta\tau Y J & (1 - 2\Delta\tau)I \end{bmatrix}, \quad (32)$$

where I denotes the $N \times N$ identity matrix. We name this map the W4 map.

Lemma 2. Suppose there exists a complete set of eigenvectors $\mathbf{v}_i \in \mathbb{R}^N$ of the matrix W and let P be an $N \times N$ matrix composed of $\mathbf{v}_i$ and λ_i be the eigenvalues corresponding to $\mathbf{v}_i$, then the error $\mathbf{e}_n^{(z)}$ converges to the zero vector if the maximum eigenvalue satisfies $|\lambda_{\max}| < 1$.

Proof. The matrix W can be decomposed as $W = P \Lambda P^{-1}$ in terms of the matrix $P := [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_N]$ and the diagonal matrix $\Lambda := \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_N]$. Then the error propagates as

$$\mathbf{e}_{n+1} = P \Lambda P^{-1} \mathbf{e}_n = (P \Lambda P^{-1})^{n+1} \mathbf{e}_0 = P \Lambda^{n+1} P^{-1} \mathbf{e}_0 \quad (33)$$

and converges to the zero vector if the maximum eigenvalue $|\lambda_{\max}|$ is less than unity. $\square$

Our problem is then reduced to finding the eigenvalues of the matrix W in Eq. (32).

Lemma 3. Suppose A, B, C, D are $N \times N$ matrices and $M := \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is a $2N \times 2N$ matrix. If A and B commute with each other and A is invertible, then the determinant of the matrix M is given as

$$\det[M] = \det[DA - CB]. \quad (34)$$

Proof. The following identity for a block matrix holds [19]:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I & -B \\ O & A \end{bmatrix} = \begin{bmatrix} A & BA - AB \\ C & DA - CB \end{bmatrix}, \quad (35)$$

where O denotes the $N \times N$ zero matrix. Since A and B commute with each other, the upper-right block vanishes. The determinant of the identity (35) yields

$$\det[M] \det[A] = \det[A] \det[DA - CB]. \quad (36)$$

Since $\det[A] \neq 0$, the last equation is reduced to

$$\det[M] = \det[DA - CB]. \quad \square \quad (37)$$

Proposition 2. Suppose there exists a solution of Eq. (1) and that $\mathbf{F} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is Lipschitz continuous near the solution $\mathbf{x}^*$ and let X and Y be $N \times N$ real nonsingular matrices, then the iteration map (30) with $Y = X^{-1} J^{-1}$ and $0 < \Delta\tau < 1$ yields a series of vectors $\mathbf{x}_n$ that converges to the solution $\mathbf{x}^*$ from an arbitrary initial guess $\mathbf{x}_0$ sufficiently close to the solution.

Proof. Let λ_W be an eigenvalue of the matrix W . Then the following equation is satisfied:

$$\det[W - \lambda_W I] = \det[(1 - \lambda_W)(1 - 2\Delta\tau - \lambda_W)I + \Delta\tau^2 Y J X] = 0, \quad (38)$$

where Lemma 3 is applied. Under the given condition $Y = X^{-1} J^{-1}$, Eq. (38) is reduced to

Table 2

The same as Table 1, but the performance of the W4 method is included. The convergence criterion is the same: $|f(x)| < 10^{-6}$. The time interval is set to $\Delta\tau = 0.5$ for the W4 method. The symbol ∞ implies again that no solution is obtained within 10^4 iteration steps.

Method \ x_i	-3.0	-2.5	-2.0	-1.5	-1.0	-0.5	0	0.5	1.0	1.5	2.0	2.5	3.0
NR	∞	∞	∞	4	5	4	3	2	4	8	4	4	3
DN	25	∞	41	20	19	20	19	15	18	19	17	19	18
W4	1434	33	70	22	25	26	25	20	22	28	30	25	24

$$\det[W - \lambda_W I] = \det[(1 - \Delta\tau - \lambda_W)^2 I] = 0, \quad (39)$$

leading to $\lambda_W = 1 - \Delta\tau$. The iteration (30) hence converges to the solution $\mathbf{x} = \mathbf{x}^*$ and $\mathbf{p} = 0$ as long as $0 < \Delta\tau < 1$, according to Lemma 2. $\square$

The Proposition implies the local convergence of our formulation, Eqs. (30a) and (30b). Note that Eqs. (29) are combined into the following second-order differential equations:

$$\ddot{\mathbf{x}} + 2\dot{\mathbf{x}} - \dot{X}X^{-1}\dot{\mathbf{x}} + XY\mathbf{F}(\mathbf{x}) = 0. \quad (40)$$

It is apparent then that $M_1 = 2I - \dot{X}X^{-1}$ and $M_2 = XY = J^{-1}$; the third term, which may be regarded as a part of the friction term, does not vanish unless the matrix X is constant, which means that the “damping factor” is in general time-dependent, a key property for the local convergence in our formulation and completely different from the NR or DN method.

3.4. UDL decomposition

Next the choice of the preconditioners X and Y is discussed. Although there are several options, one of the most useful decompositions may be the following.

Definition 1. Suppose the $N \times N$ Jacobian matrix is decomposed into $J := UDL$ where U, D and L are $N \times N$ upper triangular, diagonal and lower triangular matrices, respectively. In our formulation they are employed to give $X = L^{-1}$ and $Y = D^{-1}U^{-1}$.

Equations. (29a) and (29b) then become as follows

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta\tau L_n^{-1} \mathbf{p}_n, \quad (41a)$$

$$\mathbf{p}_{n+1} = (1 - 2\Delta\tau) \mathbf{p}_n - \Delta\tau D_n^{-1} U_n^{-1} \mathbf{F}(\mathbf{x}_n), \quad (41b)$$

where L_n^{-1}, D_n^{-1} and U_n^{-1} are the inverted matrices evaluated at $\mathbf{x}_n$: e.g., $L_n^{-1} := L^{-1}(\mathbf{x}_n)$. As mentioned in Sec. 3.3, the eigenvalue of the corresponding matrix W in Eq. (32) in the close vicinity of the solution is $1 - \Delta\tau$ to the first order.

4. Results

Here we present the results of applications of our new method, referred to as the W4 method, to some sample problems, comparing its performance with those of the NR and DN methods.

4.1. Example 1

We will return to the very first problem for a single variable given in Eq. (2):

$$f(x) = \arctan(x) + \sin(x) - 1 = 0.$$

The iteration map for this problem in the W4 method is given as

$$x_{n+1} = x_n + \Delta\tau p_n, \quad (42a)$$

$$p_{n+1} = (1 - 2\Delta\tau) p_n - \Delta\tau \frac{\arctan(x_n) + \sin(x_n) - 1}{\frac{1}{1+x_n^2} + \cos(x_n)}, \quad (42b)$$

where $0 < \Delta\tau < 1$. Table 2 summarizes the performance of the W4 method together with those of the NR and DN methods. Although the number of iteration steps is larger than that of the DN method, the W4 method successfully finds the solution where the other two methods fail. The inertia term or the second derivative term, which is absent in the NR or DN method, is responsible for the “slower but more robust” convergence in our method. In the case with $x_i = -2.5$ this effect is particularly remarkable. It is somewhat expected because Theorem 2 guarantees that there does not exist any permanent oscillation in the continuous time-evolution equation.

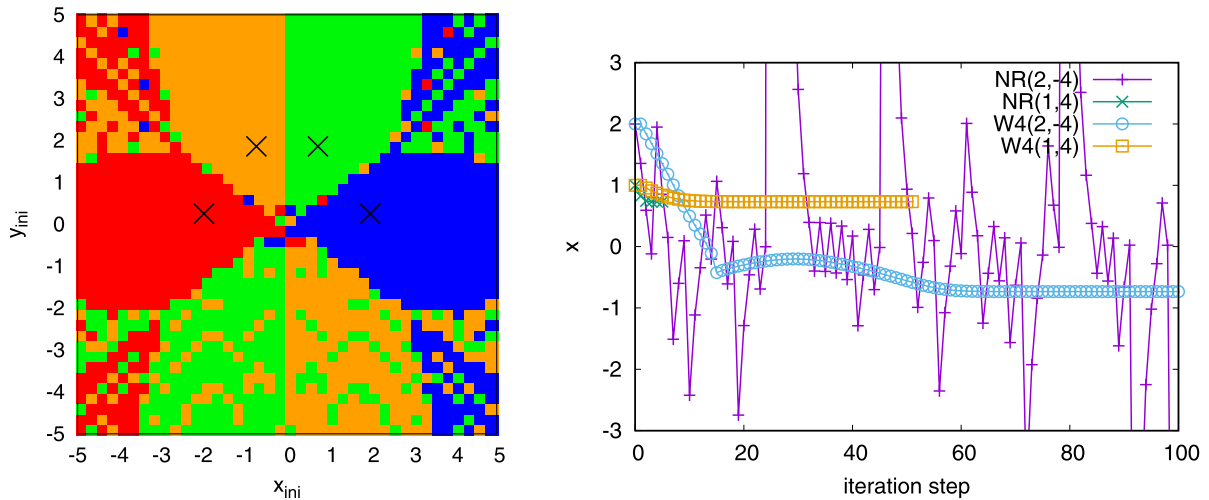


Fig. 3. (Left) the Newton basin for the W4 method with $X = L^{-1}$, $Y = D^{-1}U^{-1}$ and $\Delta\tau = 0.5$. The red, orange, green and blue colors correspond to the initial conditions to reach one of the four solutions of Eq. (5) indicated with the black crosses as in Fig. 2. (Right) the sequences produced by the iteration maps of the NR and W4 methods for the two initial conditions: $(x_0, y_0) = (1, 4)$ and $(2, -4)$.

4.2. Example 2

Next, the multi-variable problem given in Eqs. (5) is considered:

$$F_1(x, y) = x^2 + y^2 - 4 = 0,$$

$$F_2(x, y) = x^2 y - 1 = 0.$$

The W4 method with the UDL decomposition presented in Sec. 3.4 gives the following iteration map for this problem:

$$x_{n+1} = x_n + \Delta\tau p_n, \quad (44a)$$

$$y_{n+1} = y_n + \Delta\tau \left[-\frac{2y_n}{x_n} p_n + q_n \right], \quad (44b)$$

$$p_{n+1} = (1 - 2\Delta\tau) p_n - \Delta\tau \left[\frac{x_n}{2(x_n^2 - 2y_n^2)} F_1 - \frac{y_n}{x_n(x_n^2 - 2y_n^2)} F_2 \right], \quad (44c)$$

$$q_{n+1} = (1 - 2\Delta\tau) q_n - \frac{\Delta\tau}{x_n^2} F_2, \quad (44d)$$

in which $\mathbf{x} = (x \ y)^T$ and $\mathbf{p} = (p \ q)^T$.

The left panel of Fig. 3 displays the initial-guess dependence of the W4 method with $X = L^{-1}$, $Y = D^{-1}U^{-1}$ and $\Delta\tau = 0.5$. Four colors (red, orange, green and blue) correspond to the four different solutions just as in Fig. 2. It is first noted that the local convergence is observed also for the W4 method, more or less in a similar way to the NR method as expected. What is most remarkable, however, is the fact that the black region in Fig. 2, where the NR method fails to find a solution within 1000 iteration steps, disappears completely in our new method within the initial value space that has been explored. The right panel of Fig. 3 compares the sequences that the iteration maps of the two methods produce for the same initial conditions, $(x, y) = (1, 4)$ or $(2, -4)$. It is apparent that the NR method immediately finds the solution for the former whereas it fails to obtain a solution for the latter. In fact the sequence is oscillatory for the latter case and there is no hint of convergence at least up to 1000 iterations. The W4 method, on the other hand, always manages to find a solution although it takes more iterations to reach it, compared with the NR method when it does find a solution.

4.3. Example 3

The last problem is new and given by the following equations:

$$F_1(x, y) = x^2 - y^2 - 4x + 6 = 0, \quad (45a)$$

$$F_2(x, y) = 2xy + 4y - 2 = 0, \quad (45b)$$

which have actually two solutions at

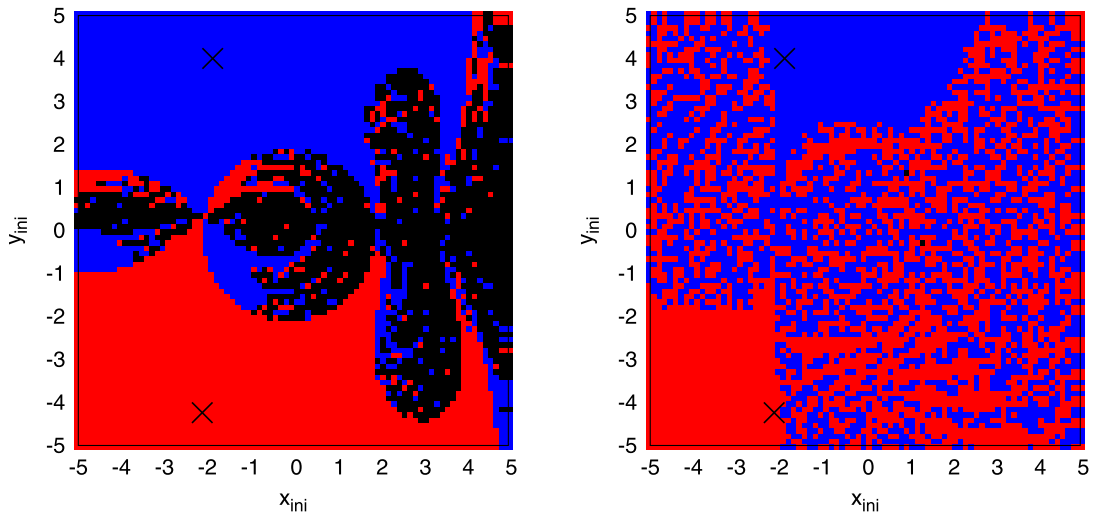


Fig. 4. The Newton basins for the NR method (left) and the W4 method with $X = D^{-1}L^{-1}$, $Y = U^{-1}$ and $\Delta\tau = 0.5$ (right). The red and blue colors correspond to the initial conditions that obtain one of the two solutions of Eq. (45) indicated with the black crosses, while the black color means that no solution is obtained within 1000 iteration steps.

$$(x^*, y^*) \approx (-1.7505169, 4.0082886), \quad (-2.2244718, -4.4549031). \quad (46)$$

The Jacobian for this problem is given as

$$J = \begin{bmatrix} 2(x-2) & -2y \\ 2y & 2(x+2) \end{bmatrix}. \quad (47)$$

(i) The iteration maps for the NR or DN method are written as

$$x_{n+1} = x_n - \frac{\Delta\tau}{2(x_n^2 + y_n^2 - 4)} \left[(x_n + 2)F_1 + y_n F_2 \right], \quad (48a)$$

$$y_{n+1} = y_n - \frac{\Delta\tau}{2(x_n^2 + y_n^2 - 4)} \left[-y_n F_1 + (x_n - 2)F_2 \right], \quad (48b)$$

where x and y are defined as $\mathbf{x} = (x \ y)^T$.

(ii) In the W4 method with the UDL decomposition, the following iteration map is solved for $\mathbf{x} = (x \ y)^T$ and $\mathbf{p} = (p \ q)^T$:

$$x_{n+1} = x_n + \Delta\tau p_n, \quad (49a)$$

$$y_{n+1} = y_n + \Delta\tau \left[\frac{y_n}{x_n + 2} p_n + q_n \right], \quad (49b)$$

$$p_{n+1} = (1 - 2\Delta\tau) p_n - \frac{\Delta\tau (x_n + 2)}{2(x_n^2 + y_n^2 - 4)} \left[F_1 + \frac{y_n}{x_n + 2} F_2 \right], \quad (49c)$$

$$q_{n+1} = (1 - 2\Delta\tau) q_n - \frac{\Delta\tau}{2(x_n + 2)} F_2. \quad (49d)$$

Fig. 4 shows the Newton basins for the iteration map (i) of the NR method (left) and for the iteration map (ii) of the W4 method with $X = L^{-1}$, $Y = D^{-1}U^{-1}$ and $\Delta\tau = 0.5$ (right). The red and blue colors correspond to the initial conditions that converge to one of the solutions indicated with the black crosses while the black dots again imply that no solution is reached within 1000 iteration steps. It is found that the blue and red regions rather clearly divided, clustering around each solution they converge to. In the middle region, however, the blue and red dots appear rather randomly and black dots show up frequently. On the other hand, the W4 method always reaches one of the solutions although it is unpredictable which one is reached, except in the close vicinity of the solutions, where the local convergence is guaranteed.

4.4. Global convergence

In this section, we elucidate the difference between the NR or DN method and the W4 method, using the following two-variable problem,

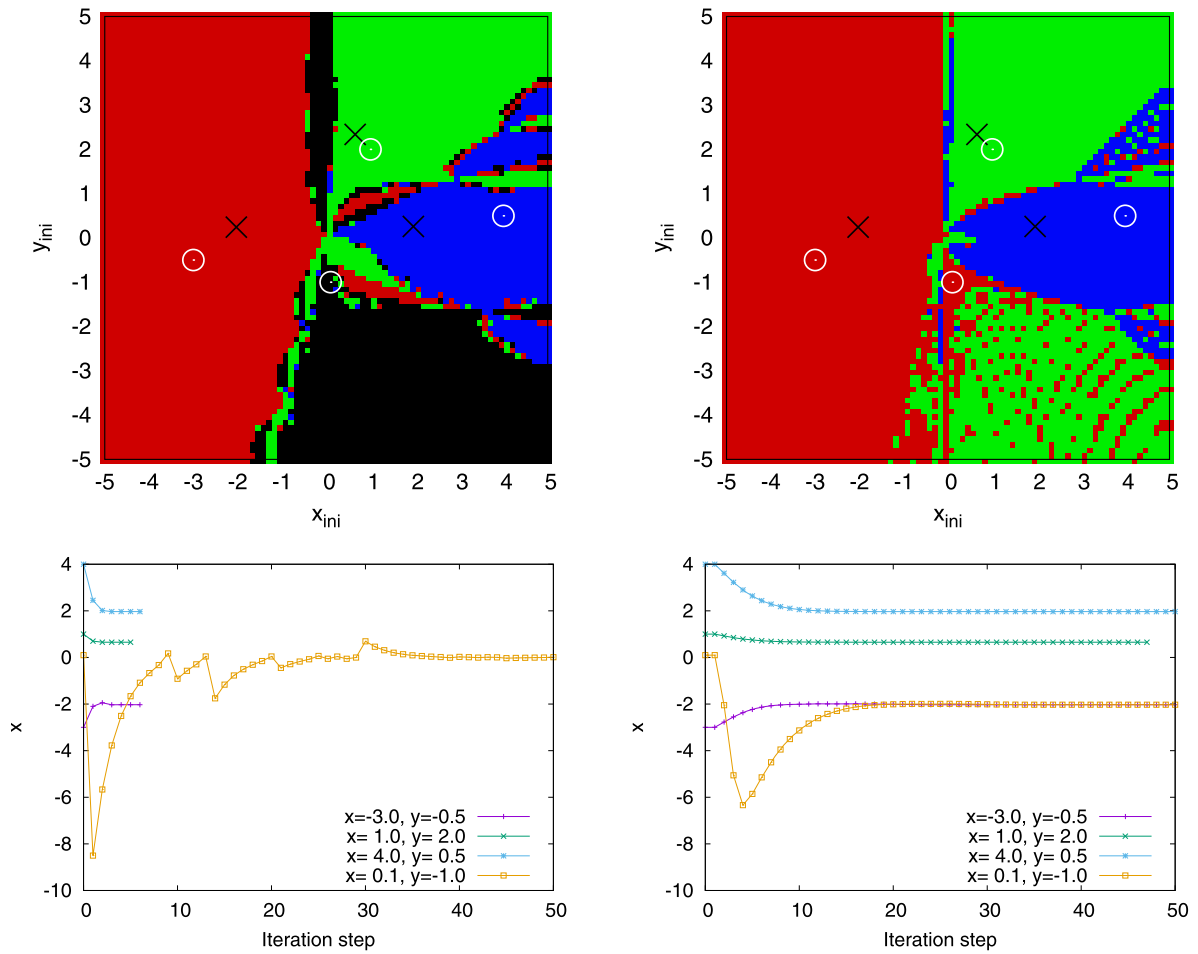


Fig. 5. The initial-guess dependence for the NR method (upper left) and the W4 method with $X = P$, $Y = \Lambda^{-1}P^{-1}$ and $\Delta\tau = 0.5$ (upper right). The red, green and blue correspond to one of the three solutions of Eq. (50), which are shown by the black crosses, while the black means the method fails to find any solution within 1000 iterations. The white circles are the initial guesses chosen for the lower evolutions from these initial guesses by the NR method (lower left) and the W4 method (lower right).

$$F_1(x, y) = x^2 + xy^2 - 4 = 0, \quad (50a)$$

$$F_2(x, y) = x^2y - 1 = 0, \quad (50b)$$

which is somewhat similar to Example 2 in Sec. 4.2 and has three solutions at

$$(x^*, y^*) \approx (-2.0296789, 0.24274223), \quad (1.9668697, 0.25849302), \quad (0.65417501, 2.3367492). \quad (51)$$

The Jacobian matrix for this system is given as

$$J = \begin{bmatrix} 2x + y^2 & 2xy \\ 2xy & x^2 \end{bmatrix}. \quad (52)$$

Note that this is a 2×2 symmetric matrix and has two real eigenvalues $\lambda_{\pm}$:

$$\lambda^{\pm} = \frac{1}{2} \left(A \pm \sqrt{A^2 - 4B} \right), \quad (53)$$

where $A := \text{tr } J = x^2 + 2x + y^2$ and $B := \det J = x^2(2x - 3y^2)$. Then the corresponding eigenvectors are given as

$$\mathbf{v}^{\pm} = \left[\frac{2xy}{a^{\pm}} \frac{\lambda_{\pm} - 2x - y^2}{a^{\pm}} \right]^T, \quad a^{\pm} := \sqrt{4x^2y^2 + (\lambda^{\pm} - 2x - y^2)^2}. \quad (54)$$

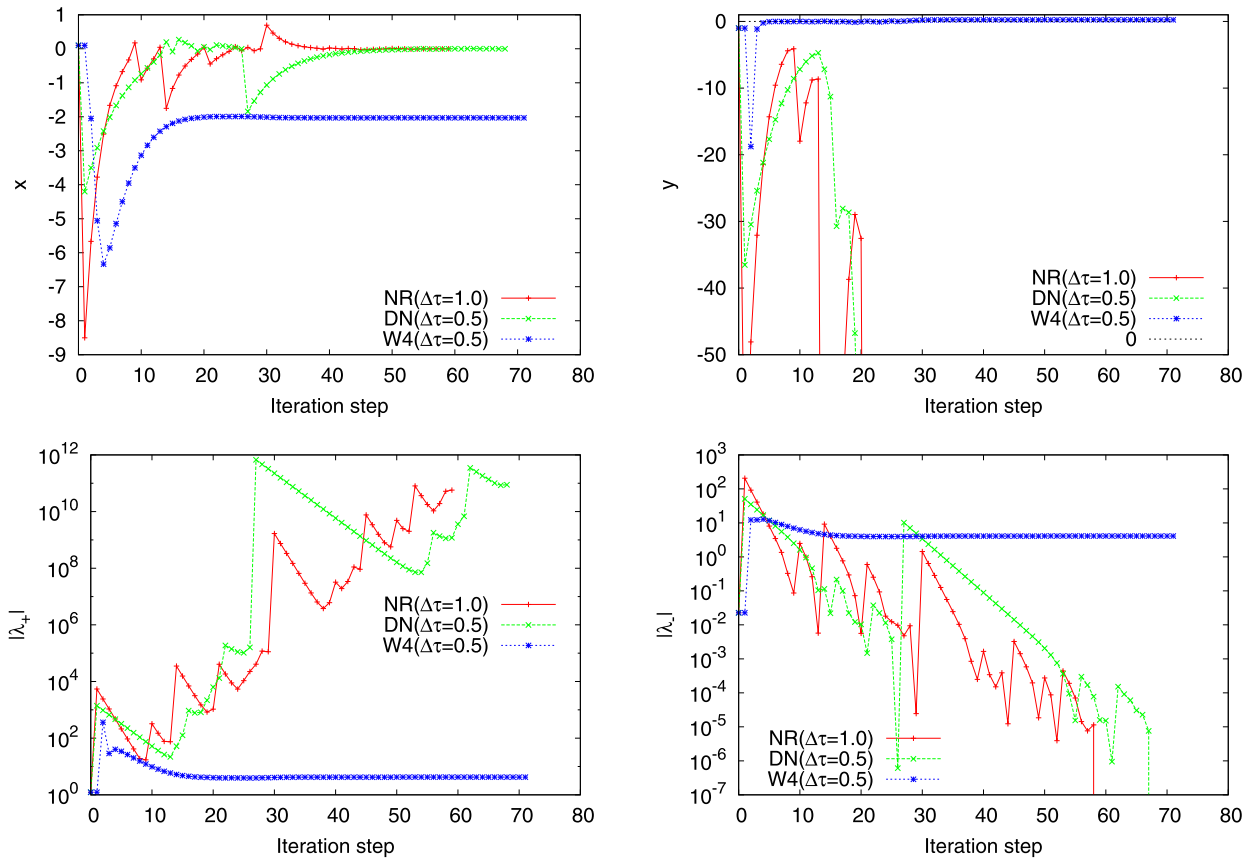


Fig. 6. The evolutionary sequences of x (upper left), y (upper right), $|\lambda_+|$ (lower left) and $|\lambda_-|$ (lower right) generated by the iteration maps of the NR, DN and W4 methods. The maximum and minimum eigenvalues of the Jacobian matrix are denoted by λ_+ and λ_- , respectively.

The iteration map for the DN method can be expressed in terms of these eigenvalues and eigenvectors as

$$\mathbf{x}_{n+1} = \mathbf{x}_n - P_n \Lambda_n^{-1} P_n^{-1} \mathbf{F}'(\mathbf{x}_n) \Delta \tau, \quad (55)$$

where $P_n = Q_n / \det(Q_n)$, $Q_n := [\mathbf{v}_n^+ \ \mathbf{v}_n^-]$, and $\Lambda_n^{-1} = \text{diag} \left[\frac{1}{\lambda_n^+} \ \frac{1}{\lambda_n^-} \right]$ for nonzero λ_n^- and $\Lambda_n^{-1} = \text{diag} \left[\frac{1}{\lambda_n^+} \ 0 \right]$ for $\lambda_n^- = 0$, all evaluated at the n -th iteration step. Note that the NR method is realized when $\Delta \tau = 1$ in Eq. (55) with nonzero λ_n^- . On the other hand, the iteration map for the W4 method is given as

$$\mathbf{x}_{n+1} = \mathbf{x}_n + P_n \mathbf{p}_n \Delta \tau, \quad (56a)$$

$$\mathbf{p}_{n+1} = (1 - 2\Delta \tau) \mathbf{p}_n - \Lambda_n^{-1} P_n^{-1} \mathbf{F}(\mathbf{x}_n) \Delta \tau. \quad (56b)$$

Note that the determinant of P_n is unity and the eigenvalues and eigenvectors change after each iteration in general.

The upper panels of Fig. 5 exhibit the Newton basins for the NR (left) and W4 (right) methods. The red, blue, and green colors correspond to the initial conditions that produce one of the three solutions indicated with the black crosses whereas the black points indicate that no solution is reached within 1000 iterations. It is clear again that the W4 method always finds some solution in sharp contrast to the NR method, which sometimes fails to reach any of the solutions. The lower panels of Fig. 5 show the specific sequences starting from the white circles displayed in the upper panels. The NR method reaches a solution very quickly within 10 iterations normally whenever it is successful. This happens usually when the initial guess is close to the solution. Otherwise it can fail, though. On the other hand, the W4 method needs more iterations but always manages to find some solution as mentioned. In the following, we investigate the possible cause of this difference more in detail, picking up the case for the initial guess $(x_i, y_i) = (0.1, -1.0)$, where the NR method fails.

The upper panels of Fig. 6 display the three evolutions of x (left) and y (right), respectively, by the NR method, the DN method with $\Delta \tau = 0.5$, and the W4 method with $\Delta \tau = 0.5$. One can see that (x_n, y_n) go to $(0, -\infty)$ as $n \rightarrow \infty$ both for the NR and DN methods. The lower panels show the evolutions of absolute values of the maximum (λ_+ , left) and minimum (λ_- , right) eigenvalues. It is evident that the ratio of $|\lambda_-/\lambda_+|$ tends to zero again both in the NR and DN methods. In what follows, a particular attention is paid to this behavior of λ_-/λ_+ and the situation when $|\lambda_-/\lambda_+|$ becomes quite small at $x \sim 0$ is analyzed.

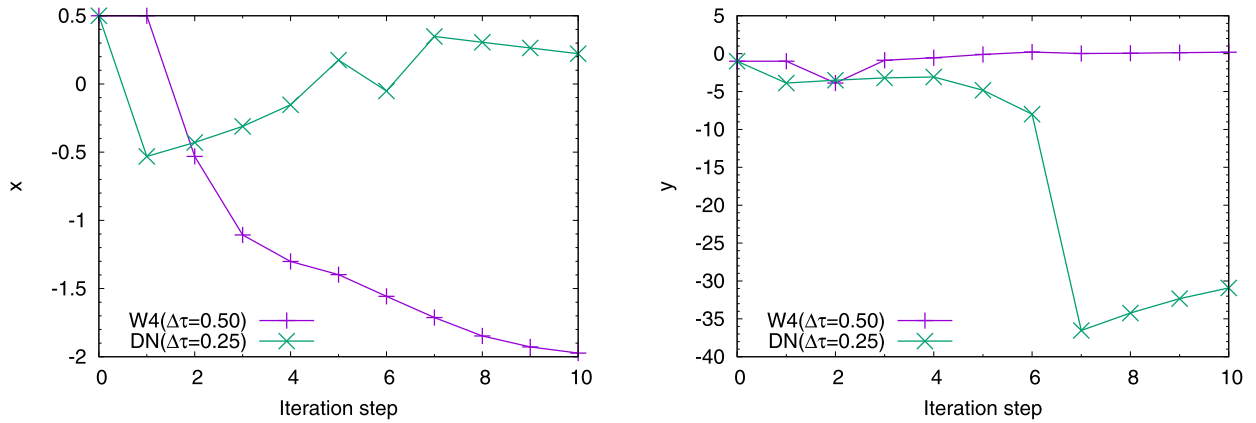


Fig. 7. The time evolutions of x (left) and y (right) for the DN method with $\Delta\tau = 0.25$ and the W4 method with $\Delta\tau = 0.5$.

Lemma 4. Let a complete set of eigenvectors of the symmetric Jacobian matrix J_n at the n -th iteration step be $\mathbf{v}_n^\pm \in \mathbb{R}^2$ and let P_n be the 2×2 orthogonal matrix composed of $\mathbf{v}_n^\pm$ and $\lambda_n^\pm$ be the corresponding eigenvalues, then the increment $\mathbf{x}_{n+1} - \mathbf{x}_n$ in Eq. (55) at this step tends to be aligned with $\mathbf{v}_n^-$ if $|\lambda_n^-/\lambda_n^+| \rightarrow 0$ as $n \rightarrow \infty$.

Proof. The iteration map (55) with $\Delta\tau = 0.5$ gives the increment in $\mathbf{x}$ at the n -th step in terms of the eigenvectors as

$$\mathbf{x}_{n+1} - \mathbf{x}_n = -\frac{1}{2}P_n\Lambda_n^{-1}P_n^{-1}\mathbf{F}(\mathbf{x}_n) = -\frac{1}{2}\frac{c_n^+}{\lambda_n^+}\mathbf{v}_n^+ - \frac{1}{2}\frac{c_n^-}{\lambda_n^-}\mathbf{v}_n^- = -\frac{1}{2\lambda_n^-}\left(\frac{\lambda_n^-}{\lambda_n^+}c_n^+\mathbf{v}_n^+ + c_n^-\mathbf{v}_n^-\right) \rightarrow -\frac{1}{2}\frac{c_n^-}{\lambda_n^-}\mathbf{v}_n^-, \quad (57)$$

where $c_n^\pm := (\mathbf{v}_n^\pm)^T \mathbf{F}(\mathbf{x}_n)$ is introduced and the limit $|\lambda_n^-/\lambda_n^+| \rightarrow 0$ is taken. $\square$

This Lemma explains the behavior of the sequences produced by the iteration map of the DN method. In fact, Taylor-expanding the eigenvalues and eigenvectors as well as the coefficients $c^\pm$ in the neighborhood of $x = 0$,

$$\lambda^+ = y^2 + 2x + 4x^2 + \mathcal{O}(x^3), \quad \lambda^- = -3x^2 + \mathcal{O}(x^3), \quad (58)$$

$$\mathbf{v}^+ = \begin{pmatrix} -1 + \frac{2x^2}{y^2} + \mathcal{O}(x^3) & -\frac{2x}{y} + \frac{4x^2}{y^2} + \mathcal{O}(x^3) \end{pmatrix}^T, \quad (59)$$

$$\mathbf{v}^- = \begin{pmatrix} \frac{2x}{y} - \frac{4x^2}{y^2} + \mathcal{O}(x^3) & -1 + \frac{2x^2}{y^2} + \mathcal{O}(x^3) \end{pmatrix}^T, \quad (60)$$

$$c^+ = 4 + \left(-y^2 + \frac{2}{y}\right)x + \mathcal{O}(x^2), \quad c^- = 1 - \frac{8x}{y} + \mathcal{O}(x^2), \quad (61)$$

one realizes that once the sequence enters the region with $y < 0$, it heads to $x = 0$ and $y = -\infty$. This is also understood from the fact that $\mathbf{v}^+$ is the only way out of this region.

Lemma 5. Let a complete set of eigenvectors of the symmetric Jacobian matrix J_{n-1} at the $(n-1)$ -th iteration step be $\mathbf{v}_{n-1}^\pm \in \mathbb{R}^2$ and let P_{n-1} be the 2×2 orthogonal matrix composed of $\mathbf{v}_{n-1}^\pm$ and $\lambda_{n-1}^\pm$ be the corresponding eigenvalues, then in the iteration map of the W4 method with $\Delta\tau = 0.5$ the increment in $\mathbf{x}$ does not tend to be aligned completely with $\mathbf{v}_{n-1}^-$ but depends also on $\mathbf{v}_{n-1}^+$ as $|\lambda_{n-1}^-/\lambda_{n-1}^+| \rightarrow 0$ if $\mathbf{v}_n^-$ is not parallel to $\mathbf{v}_{n-1}^-$.

Proof. The iteration map (56) of the W4 method with $\Delta\tau = 0.5$ gives

$$\mathbf{x}_{n+1} - \mathbf{x}_n = -\frac{1}{2}P_n\mathbf{p}_n = -\frac{1}{4}P_n\Lambda_{n-1}^{-1}P_{n-1}^{-1}\mathbf{F}(\mathbf{x}_{n-1}) = -\frac{1}{4\lambda_{n-1}^-}\left(\frac{\lambda_{n-1}^-}{\lambda_{n-1}^+}c_{n-1}^+\mathbf{v}_n^+ + c_{n-1}^-\mathbf{v}_n^-\right) \rightarrow -\frac{1}{4}\frac{c_{n-1}^-}{\lambda_{n-1}^-}\mathbf{v}_n^-. \quad (62)$$

One can recognize the same alignment of the increment with the eigenvector $\mathbf{v}_n^-$ at the n -th iteration step. Note, however, that the singularity in the Jacobian matrix, or $|\lambda_{n-1}^-/\lambda_{n-1}^+| \approx 0$, is assumed to emerge at the $(n-1)$ -th iteration step this time. The above equation implies that the alignment occurs one step later. Since $\mathbf{v}_n^-$ can be expanded in terms of the eigenvectors at the $(n-1)$ -th iteration step as

$$\mathbf{v}_n^- = a^+\mathbf{v}_{n-1}^+ + a^-\mathbf{v}_{n-1}^-, \quad (63)$$

where $a^\pm$ are certain constants, the increment in $\mathbf{x}_n$ can be rewritten as

$$\mathbf{x}_{n+1} - \mathbf{x}_n = -\frac{a^+ c_{n-1}^-}{4\lambda_{n-1}^-} \mathbf{v}_{n-1}^+ - \frac{a^- c_{n-1}^-}{4\lambda_{n-1}^-} \mathbf{v}_{n-1}^-, \quad (64)$$

and hence includes the contribution from $\mathbf{v}_{n-1}^+$ unless a^+ is exactly zero, $a^+ = 0$. $\square$

The important thing in the Lemma above is that the singularity in the Jacobian matrix at a certain iteration step affects the direction of the sequence at the next step but it is $\mathbf{v}_{n-1}^+$ that is the direction of crucial importance so that the sequence has a chance to get out of the vicinity of $x = 0$, in which the sequence of the DN method is trapped.

Fig. 7 shows the specific evolutions of x (left) and y (right) for the iteration maps of the DN method with $\Delta\tau = 0.25$ and of the W4 method with $\Delta\tau = 0.5$, both starting from the same initial condition $(x, y) = (0.5, -1.0)$ (and $(p, q) = (0, 0)$ in addition for the latter). Note that (x_1, y_1) at the first step in the DN method is identical to (x_2, y_2) in the W4 method but the following evolutions are different from each other. In fact, x_2 in the DN method gets closer to $x = 0$ whereas x_3 in the W4 method is farther away, since the momentum gained in the previous steps prevents it from changing the direction of motion in the latter method.

5. Applications to an astrophysical problem

In this section, we demonstrate the applicability of our method to much higher-order multi-variable problems as we frequently encounter in physics problems. We take an example from astrophysics here. The structure of a spherically symmetric static star is determined by the balance between gravity and pressure [21,22]. We need to employ general relativity for gravity if the star is compact and massive. Then the basic equations are the Euler equation for matter and the Einstein equation for gravity, which are given in this context as

$$\frac{8\pi G}{c^2} r^2 \rho(r) - \frac{2r}{g(r)^3} \frac{dg}{dr}(r) - 1 + \frac{1}{g(r)^2} = 0, \quad (65)$$

$$\frac{8\pi G}{c^4} r^2 P(r) + \frac{2r}{g(r)^2} \frac{1}{P(r) + \rho(r)c^2} \frac{dP}{dr}(r) + 1 - \frac{1}{g(r)^2} = 0, \quad (66)$$

where G and c are the gravitational constant and the speed of light, respectively; the functions to be solved, $P(r)$, $\rho(r)$ and $g(r)$, are the pressure, density, and metric component, respectively; all of them are functions of the radius r alone. The above equations are equivalent to the well-known Tolman-Oppenheimer-Volkoff equation in fact [21,22]. The system of equations is closed by specifying the equation of state (EOS) which is assumed here for simplicity as

$$P = K \rho^\Gamma, \quad (67)$$

where K and Γ are constants. In the following, the geometrical unit $c = G = 1$ as well as $K = 1$ are adopted.

Introducing further the non-dimensional radius $\xi = r/R$, where R is the stellar radius, we rewrite the basic equations as follows:

$$8\pi R^2 \xi^2 \rho(\xi) - \frac{2\xi}{g(\xi)^3} \frac{dg}{d\xi}(\xi) - 1 + \frac{1}{g(\xi)^2} = 0, \quad (68)$$

$$8\pi R^2 \xi^2 P(\xi) + \frac{2\xi}{g(\xi)^2} \frac{1}{P(\xi) + \rho(\xi)} \frac{dP}{d\xi}(\xi) + 1 - \frac{1}{g(\xi)^2} = 0. \quad (69)$$

The second-order discretization of these equations on the equally-spaced grid in ξ ($0 \leq \xi \leq 1$) gives the following nonlinear system of equations:

$$F_j^{(g)} := 8\pi R^2 \xi_{j-1/2}^2 \rho_{j-1/2} - \frac{2\xi_{j-1/2}}{g_{j-1/2}^3} \frac{g_j - g_{j-1}}{\Delta\xi} - 1 + g_{j-1/2}^{-2} = 0, \quad (70)$$

$$F_j^{(\rho)} := 8\pi R^2 \xi_{j-1/2}^2 P_{j-1/2} + \frac{2\xi_{j-1/2}}{g_{j-1/2}^2} \frac{1}{P_{j-1/2} + \rho_{j-1/2}} \frac{P_j - P_{j-1}}{\Delta\xi} + 1 - g_{j-1/2}^{-2} = 0, \quad (71)$$

where j runs from 1 to N_s , the number of grid points; $\Delta\xi = 1/N_s$; the variables with the subscript $j - 1/2$ denote the average between adjacent points, e.g., $\rho_{j-1/2} := (\rho_j + \rho_{j-1})/2$. The boundary conditions need to be imposed: (i) the density at the surface should vanish, $\rho_{N_s} = 0$ and (ii) the spacetime is flat at the center of the star and becomes the Schwarzschild spacetime at the surface of the star, $g_0 = 1$, $g_{N_s} = 1/\sqrt{1 - 2M/R}$, where M is the stellar (gravitational) mass.

The variables and equations to be solved are summarized as

$$\mathbf{x} = (\rho_0, \rho_1, \dots, \rho_{N_s-1}, g_1, g_2, \dots, g_{N_s-1}, R), \quad (72)$$

$$\mathbf{F} = (F_1^{(\rho)}, F_2^{(\rho)}, \dots, F_{N_s}^{(\rho)}, F_1^{(g)}, F_2^{(g)}, \dots, F_{N_s-1}^{(g)}, F_{N_s}^{(g)}) = \mathbf{0}. \quad (73)$$

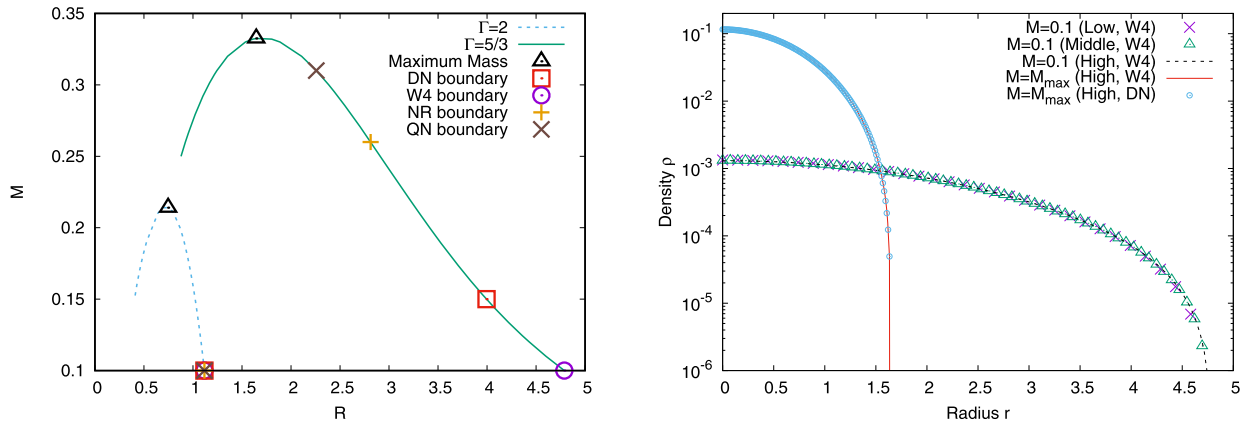


Fig. 8. (Left) The mass-radius relations for the relativistic stars with $\Gamma = 2$ (dashed line) and $\Gamma = 5/3$ (solid line). Triangles denote the stars with the maximum mass in each sequence. Pluses, squares, crosses, and circles indicate the lower limit of the initial configurations, down to which the NR method, the DN method, a quasi-Newton (QN) method, and the W4 method, respectively, can find the solution with the maximum mass from these initial guesses with wrong masses. (Right) The density profiles of the stars with $M = 0.1$ and the maximum mass. For the former, the radius for three different resolutions, i.e., $N_s = 32$ (Low), $N_s = 64$ (Middle), $N_s = 128$ (High) are shown with crosses, triangles and a dashed line, respectively. For the maximum-mass case, only the high-resolution result obtained with the DN method is also presented. Γ is set to $5/3$ in all the cases.

It is evident that the number of the variables, which is equal to that of the equations as it should, is $N = 2N_s$. Note that the pressure is obtained from the density as $P_j = \rho_j^\Gamma$ and is not counted as the variables to be solved; the radius of the star is unknown a priori and is obtained as the position where the density (and hence the pressure) vanishes. These equations with the boundary conditions are solved with both the DN method and the W4 method.

We display the results in Fig. 8. Note that each stellar structure is characterized by the stellar mass M and the so-called adiabatic index Γ . In the left panel we show the mass-radius relations obtained with the W4 method for $\Gamma = 2$ (solid line) and $\Gamma = 5/3$ (dashed line). The well-known fact that there is a maximum mass (marked with triangles) in each sequence is reproduced. The right panel of Fig. 8 exhibits the density profiles obtained with the W4 method for $\Gamma = 5/3$. The stellar mass is set either to 0.1 or to the maximum mass of this sequence. For the former, we deploy three grids with different resolutions: $N_s = 32$ (low resolution), $N_s = 64$ (middle resolution) and $N_s = 128$ (high resolution), whereas we use only the high-resolution grid for the latter. For comparison, the profile obtained with the DN method is also shown for the maximum mass. One can see that the numerical results are already converged at $N_s = 32$. Note also that the star with the maximum mass is much more compact than the star with the mass of 0.1, another well-known fact. The results are in good agreement between the two methods. We can hence conclude that the W4 method is successfully applied to this higher-order problem.

Now we want to quantify the global convergence of the two methods. For this purpose we choose as the initial guess the solutions with wrong masses and see up to what mass the solution with the true mass can be obtained. The results are shown in the left panel of Fig. 8. The true mass is set to the maximum mass and the wrong mass in the initial guess is changed from the maximum mass downward. For $\Gamma = 2$, both methods can find the solution from the initial guess with the wrong mass as small as $M = 0.1$ corresponding to the square and circle, respectively, in the panel. For $\Gamma = 5/3$, on the other hand, the DN method fails to find the solution below $M = 0.15$ while the W4 method successfully reaches solutions all the way down to $M = 0.1$. Note that the change in radius with mass is more remarkable for $\Gamma = 5/3$ than for $\Gamma = 2$. For reference, the results with the NR method and a quasi-Newton (QN) method [1,9] are also shown by pluses and crosses, respectively. They are even worse than the DN method in the global convergence.

6. Conclusion

We have proposed a new iterative method, which we named the W4 method, to solve nonlinear systems of equations. In this paper, we have considered a couple of representative problems, for which the Newton-Raphson method, or the NR method, fails to find any one of the solutions, and have applied the new method to them to see if any improvement is made. We have also applied the method to an astrophysical problem with many (~ 100) variables.

Our findings are listed as follows:

1. When the NR method or its modified version, the DN method, fails in these test problems, their iteration maps produce sequences, which either diverge or oscillate as shown in Fig. 3.
2. The W4 method, which introduces the second time derivative term, can be regarded as an extension of the NR method as argued in Sec. 3.2 and Sec. 3.3.
3. The W4 method has the same local convergence as the one that the NR method is well known to possess, as demonstrated in Sec. 3.3.

4. In the W4 method, the convergence occurs only linearly just as in the DN method. The NR method, in contrast, has a quadratic convergence nature. This was discussed in Sec. 3.1 and Sec. 3.3.
5. In the W4 method the Newton basins that lead to no solution in the NR method are eliminated as exhibited in Figs. 2, 3 and 4. This strongly suggests that the W4 method has a better global-convergence property in multi-variable problems. This is also demonstrated in the astrophysical problem with much more variables in Sec. 5.

The computational cost of the current version of the W4 method, which employs the UDL decomposition of the Jacobian matrix, is no higher than that of the NR method because the latter also employs the same decomposition of the Jacobian matrix. We have already applied the W4 method to several other (more realistic) astrophysical problems that require numerical solutions of a large system of nonlinear equations and have found it successful as multi-dimensional root-finder [4,20,5,12]. Very encouraging as these results may look, more works are certainly needed to give theoretical and/or experimental justifications.

We hope that our new method, having the better global-convergence nature, will be very useful in many fields of science and engineering, particularly when the NR method fails to work.

Acknowledgements

We would like to thank K.-i Maeda for helpful comments. We are grateful to Y. Eriguchi for valuable comments on the preliminary draft. H. O. is grateful to Z. Pelgrims for reproducing our result. This work was supported by JSPS KAKENHI Grant Number 16K17708, 16H03986, 17K18792, 20K03953, 20K14512. R. H. was supported by JSPS overseas research fellowship No. 29-514. K. F was supported by JSPS postdoctoral fellowships No. 16J10223. H. N. was partially supported at Caltech through NSF award No. TCAN AST-1333520 and DOE SciDAC4 Grant DE-SC0018297 (subaward 00009650). S. Y. is supported by the Institute for Advanced Theoretical and Experimental Physics, Waseda University and the Waseda University Grant for Special Research Projects (project number: 2020C-273, 2021C-197, 2022C-140).

References

- [1] C.G. Broyden, A class of methods for solving nonlinear simultaneous equations, *Math. Comput.* 19 (92) (1965) 577–593.
- [2] E.W. Cheney, D.R. Kincaid, *Numerical Mathematics and Computing*, Cengage Learning, 2012.
- [3] X. Fang, Q. Ni, M. Zeng, A modified quasi-Newton method for nonlinear equations, *J. Comput. Appl. Math.* 328 (2018) 44–58.
- [4] K. Fujisawa, H. Okawa, Y. Yamamoto, S. Yamada, Effects of rotation and magnetic field on the revival of a stalled shock in supernova explosions, *Astrophys. J.* 872 (2) (2019) 155, <https://doi.org/10.3847/1538-4357/aaffdd>, arXiv:1809.04358.
- [5] R. Hirai, T. Sato, P. Podsiadlowski, A. Vigna-Gomez, I. Mandel, Formation pathway for lonely stripped-envelope supernova progenitors: implications for Cassiopeia A, *Mon. Not. R. Astron. Soc.* 499 (1) (2020) 1154–1171, <https://doi.org/10.1093/mnras/staa2898>, arXiv:2008.05076.
- [6] A.S. Householder, *The Numerical Treatment of a Single Nonlinear Equation*, McGraw-Hill, 1970.
- [7] J.L. Hueso, E. Martínez, J.R. Torregrosa, Modified newtons method for systems of nonlinear equations with singular Jacobian, *J. Comput. Appl. Math.* 224 (1) (2009) 77–83.
- [8] A. Jablonski, Monte Carlo algorithm for solving systems of non-linear equations, *J. Comput. Appl. Math.* 6 (3) (1980) 171–175.
- [9] C.T. Kelley, *Solving Nonlinear Equations with Newtons Method*, Vol. 1, Siam, 2003.
- [10] J. Kou, Y. Li, X. Wang, Efficient continuation Newton-like method for solving systems of non-linear equations, *Appl. Math. Comput.* 174 (2) (2006) 846–853.
- [11] V.K. Kyncheva, V.V. Yotov, S.I.I. ov, Convergence of Newton, Halley and Chebyshev iterative methods as methods for simultaneous determination of multiple polynomial zeros, *Appl. Numer. Math.* 112 (2017) 146–154.
- [12] H. Okawa, K. Fujisawa, N. Yasutake, M. Ogata, Y. Yamamoto, S. Yamada, A novel formulation for the evolution of relativistic rotating stars, arXiv:2204.09943, 2022.
- [13] J.M. Ortega, W.C. Rheinboldt, *Iterative Solution of Nonlinear Equations in Several Variables*, Vol. 30, Siam, 1970.
- [14] W.H. Press, S.A. Teukolsky, W.T. Vetterling, B.P. Flannery, *Numerical Recipes 3rd Edition: The Art of Scientific Computing*, 3rd edition, Cambridge University Press, USA, 2007.
- [15] H. Ramos, M.T.T. Monteiro, A new approach based on the newtons method to solve systems of nonlinear equations, *J. Comput. Appl. Math.* 318 (2017) 3–13.
- [16] H. Ramos, J. Vigo-Aguiar, The application of newtons method in vector form for solving nonlinear scalar equations where the classical Newton method fails, *J. Comput. Appl. Math.* 275 (2015) 228–237.
- [17] S. Regmi, *Optimized Iterative Methods with Applications in Diverse Disciplines*, Nova Science Publishers, 2020.
- [18] S. Schlenkrich, A. Walther, Global convergence of quasi-Newton methods based on adjoint Broyden updates, *Appl. Numer. Math.* 59 (5) (2009) 1120–1136.
- [19] J.R. Silvester, Determinants of block matrices, *Math. Gaz.* 84 (501) (2000) 460–467.
- [20] H. Suzuki, P. Gupta, H. Okawa, K.-i. Maeda, Post-Newtonian Kozai–Lidov mechanism and its effect on cumulative shift of periastron time of binary pulsar, *Mon. Not. R. Astron. Soc.* 500 (2) (2020) 1645–1665, <https://doi.org/10.1093/mnras/staa3081>, arXiv:2006.11545.
- [21] K.S. Thorne, C.W. Misner, J.A. Wheeler, *Gravitation*, Freeman, San Francisco, CA, 2000.
- [22] R.M. Wald, *General Relativity*, University of Chicago Press, 2010.
- [23] X. Wu, Note on the improvement of newtons method for system of nonlinear equations, *Appl. Math. Comput.* 189 (2) (2007) 1476–1479.